



United International University (UIU)
Dept. of Computer Science & Engineering (CSE)

Final Exam, Trimester: Summer 2025

Course Code: CSE 1111, Course Title: STRUCTURED PROGRAMMING LANGUAGE

Time: 2 hours Total Marks: 40

(Unfair means in the exam will be taken seriously as per UIU disciplinary rules)

Answer all the Questions. Numerical values in the right margin indicate full marks.

- 1 a) Write a **program** using **two user defined functions** to perform the following operations [5]
- i) The function **main()** reads three arms of a triangle from keyboard.
 - ii) The user defined function **isTriangle(int x, int y, int z)** checks whether the 3 arms can construct a valid triangle. If it is, it will return **1**, otherwise **0**. [Valid triangle condition: the sum of any two arms is greater than the third one.]
 - iii) The user defined function **areaCal(int x, int y, int z)** calculates the area of a triangle if the **isTriangle()** function returns 1, and returns area to the calling function. [Hint: $\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$, where semi-perimeter, $s=(a+b+c)/2$.]
 - iv) The **main()** function prints the area of the triangle if valid using above functions.
- b) Do **manual tracing** for the following program. Find and provide the **outputs**. Keep your work **neat and clean** with labels. [5]

```
#include<stdio.h>
int a, b;
void func2(int x, float y){
    printf("%d %f\n", x+5, y+6);
}
int func1(float x) {
    b=b*a;
    printf("%f\n", x+3);
    func2(10, 6.5);
    return b-1;
}
void main(){
    a=11;
    b=25;
    printf("%d %d\n", b, a);
    a=func1(9.5);
    func2(12, 20.0);
    printf("%d %d\n", a, b);
}
```

Program for 1 (b)

```
#include<stdio.h>
void change(int *x, int b){
    int *p;
    p=x;
    *p=*p+10;
    b=b-3;
}
int main(){
    int a=5, b=9;
    printf("%d %d\n", a, b);
    change(&a, b);
    printf("%d %d", a, b);
    return 0;
}
```

Program for 2 (b)

- 2 a) Show **manual tracing** for the following program [4]
- ```
char Str1[8] = {"\0"};
char Str2[4] = {"\0"};
int i;
strcpy(Str1, "UIU");
strcpy(Str2, "CSE");
strcat(Str1, " ");
strcat(Str1, Str2);
i = strlen(Str1);
if (strcmp(Str2, Str1)<0)
 printf("String=%s length=%d\n", Str2, strlen(Str2));
else
 printf("String=%s length=%d\n", Str1, i);
```
- b) Find **output** for the program given above at the right. [4]

3 a) Find **output** for the following program (left).

[5]

```
#include<stdio.h>
void func(int B[], int n);
void main(){
 int A[5]={10, 20, 30, 40, 50};
 int i;
 for(i=0; i<5; i++)
 printf("%d ", A[i]);
 func(A, 5);
 printf("\n");
 for(i=0; i<5; i++)
 printf("%d ", A[i]);
}
void func(int B[], int n){
 int i;
 for(i=0; i<n; i++)
 B[i] = B[i]*(i+1);
}
```

Program for 3 (a)

```
#include<stdio.h>
void main(){
 FILE *fp1;
 int i, sum;
 int num[5]={10, 20, 30, 40, 50};
 fp1= fopen("D:\\students\\dest.txt",
"w");
 sum=0;
 for(i=4; i>=0; i--){
 if(i%2==0){
 sum=sum+num[i];
 fprintf(fp1, "%d\n", num[i]);
 }
 }
 fprintf(fp1, "%d", sum);
 fclose(fp1);
}
```

Program for 3 (b)

b) What will be the **effect** of the program given above at the right?

[5]

4 Write a **program** having the structure student (name, id, marks) to perform the following operations for **n** students

[8]

- Read name, id, marks of **n** students from keyboard
- Find the minimum marks and the maximum marks holder students
- Find the average marks of **n** students
- Display a report (see a sample below) on monitor for **n** students (here, **n**=4):

|        |    |      |
|--------|----|------|
| Rahim  | 10 | 85.0 |
| Saiham | 20 | 85.4 |
| Sabera | 15 | 82.8 |
| Farhan | 18 | 80.0 |

Minimum marks holder student: Farhan 18  
Maximum marks holder student: Saiham 20  
Average: 83.3

5 Show the manual tracing for the following recursive function for a func(5) call.

[4]

```
int func(int x){
 if (x == 1)
 return 1;
 else
 return x+ func(x-1);
}
```

**OR**

Write a program to convert the uppercase letters of an input string into lowercase. Remember that you are not allowed to use any standard function for this case.

For example,

Input string: "Uiu Cse"  
Output string: "uiu cse"